

COMP.SE.140 – Docker-compose and microservices report

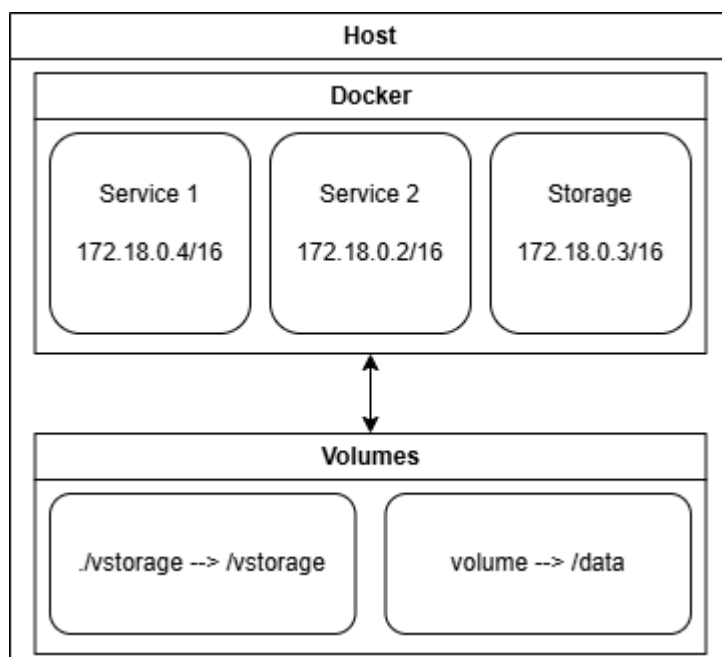
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Source code: <https://github.com/lindaormark/devops>

Basic information:

- Hardware, OS Windows 11 Home
- Docker Desktop 4.46.0 (204649)
- Docker v28.4.0
- v2.39.2-desktop.1

Diagram



Content of the status records

The services measure the space in the container's root filesystem and the uptime of the container. Container uptime is a pretty relevant measurement as it can be used to define if there's been a crash or a restart in some point, but knowledge about the container's root disk space is not very relevant. Something like memory or CPU usage would be more beneficial in keeping track of the containers.

Analysis and comparison of persistent storage solutions

Shared folder:

The shared folder is easier to set up and the logs are easy to check any time you want. It doesn't rely on Docker to work and is quite easy to backup. It's also been used much during studies until now because it's easy to see the effect of your code. The downsides are that the commands for accessing a file are different on each platform, it's not portable and it's not very secure as the container accesses the host's file system.

Storage service:

Somewhat harder to setup especially if you've never done it before but the contents are accessible and consistent across platforms, scales much better than a local file and it's more secure since the container doesn't need to access the host system. Some downsides are that it's harder to inspect, it's useless if the storage service is down and it requires instructions to clear instead of just deleting a file.

Teacher's guide to cleaning up the persistent storage

1. `docker compose down -v` (Stop services and remove volumes)
2. `rm -rf ./vstorage` (Remove local folder on Linux)
3. `docker compose up --build -d` (Restart services)

Difficult parts

I had only used Docker in one school project before this so pretty much everything was difficult, and I think my implementation is still far from perfect. I found writing the docker-compose files and Dockerfiles the most difficult task, it was hard to differentiate what parts were really necessary based on endless Googling of example files.

Main problems

It was difficult to get the services to print correctly, meaning that usually it would print to Docker's terminal but not the host's. I spent a lot of time trying to fix it and eventually did. It's harder to find where the problem is with containers than without them.